Value of Barnebys user for Skeleton Clients

Technical Documentation

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1. Executive Summary

This document describes the technical implementation of the Skeleton 34 Auction House Pipeline, covering auction data for 2025 (excluding June–July).

Objective

Expand the Olséns pilot analysis to 34 Skeleton auction houses, automating data extraction and processing to measure the value Barnebys users bring at scale.

Technical approach

The data for this analysis across four systems: Bite, Skeleton, Barnebys AWS and Azure, and auction house data.

An automated Python pipeline extracts data from three sources (Skeleton SQL Server, AWS MySQL, Azure MySQL) and loads it into BigQuery, manually extract data from Bite and loads them into BigQuery. A series of SQL scripts then process the data through three layers (Raw → Processed → Analytics), producing unified metrics across all 34 auction houses.

Scope

- 34 Skeleton auction houses
- Time period: January – December 2025 (excluding June–July)
- BigQuery dataset: barnebys-skeleton.42ah (Personal BigQuery)
- Output: Looker Studio dashboard

Relation to Olséns Pilot

This pipeline is a direct expansion of the Olséns pilot. Core SQL processing logic is inherited; key additions are the Python extraction layer, multi-auction-house handling, Skeleton fee calculation, and auction house performance tiering.

2. Business Context

Background

Same as Olséns pilot analysis. Barnebys drives traffic to Skeleton auction houses via fingerprint-based tracking (Bite). See [Olséns Pilot Technical Documentation] for full background on Skeleton, Bite, and tracking mechanism.

This pipeline extends the pilot to the Skeleton client base. Out of 46 total Skeleton clients, only 34 have bid data recorded in Bite and are included in this analysis.

The BigQuery dataset is named barnebys-skeleton.42ah for historical reasons.

Analysis scope

Parameter	Value
Auction houses	34 Skeleton clients
Time period	Jan – Dec 2025
Excluded months	June – July (source tracking issues)
BigQuery dataset	barnebys-skeleton.42ah

Data source and Extraction period

Source	Data	Extraction Period	Method
Skeleton SQL Server	Auctions, EnteredBids	Jan 2025 – Jan 2026	extract.py (Python)
AWS MySQL	Lot metadata	Nov 2024 – May 2025	extract.py (SSH tunnel)
Azure MySQL	Lot metadata	Jun 2025 – Dec 2025	extract.py (direct)
Bite BigQuery	Clicks, registrations, bids	Jan – Dec 2025	BigQuery SQL queries

Manual upload	Skeleton pricing, currency rates, AH reference	2025 annual	CSV → BigQuery

Manual reference tables

Three tables are manually maintained and uploaded to BigQuery:

Table	Content	Source
raw_skeleton_pricing	Fee structure per auction house (type, rate, min/fixed fee, currency)	Skeleton team
raw_currency_eur	2025 annual average EUR exchange rates	Google (2025 avg)
raw_auction_house	Auction house reference data (id, name, Skeleton client flag)	Internal

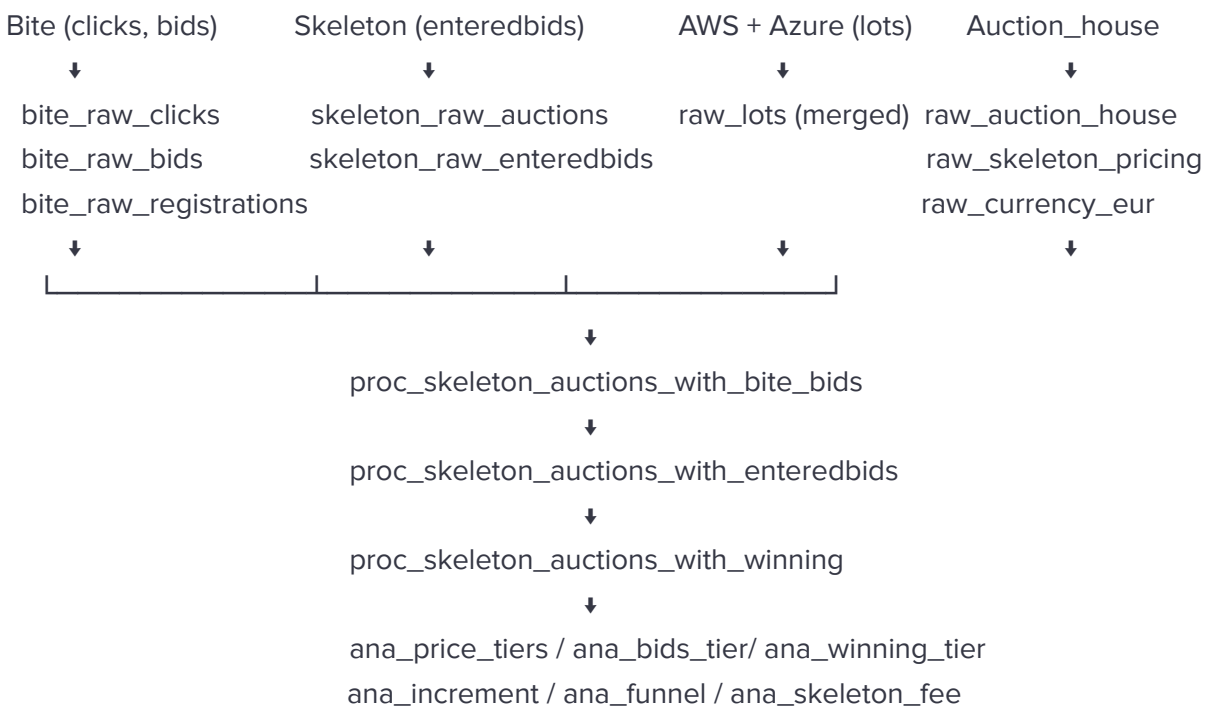
3. Data Architecture

Three-Layer Architecture

All data processing is implemented in BigQuery, following a three-layer pattern:

Layer	Purpose	Tables
Raw	Extract and upload raw data to BigQuery	bite_raw_clicks bite_raw_registrations bite_raw_bids aws_barnebys_raw_lots azure_barnebys_raw_lots raw_skeleton_auctions raw_skeleton_enteredbids raw_auction_house raw_skeleton_pricing raw_currency_eur
Processed	Clean, match, and enrich records	update_raw_bite_bids raw_bbys_lots raw_bite_bids_clean proc_skeleton_auctions_with_bite_bids proc_skeleton_auctions_with_enteredbids proc_skeleton_auctions_with_winning
Analytics	Aggregated metrics for analysis	ana_lot_price_tiers ana_bids_with_price_tier ana_winning_bids_with_price_tier ana_barnebys_increment ana_funnel_all_stats ana_funnel_excluded67_unpivoted ana_funnel_olsens_exclued67 ana_skeleton_fee

Data Flow



Key tables

Input tables:

Table	Description	Source	Record Count
bite_raw_clicks	Raw click events from Bite	Bite	283
bite_raw_registrations	User registration events	Bite	29,402
bite_raw_bids	Raw bid events from Bite	Bite	1,036,957
raw_skeleton_auctions	Auction & inventoryld records	Skeleton SQL Server	203,481
raw_skeleton_enteredbids	All entered bid records	Skeleton SQL Server	1,069,364
raw_aws_barnebys_lots	Lot data from AWS	AWS MySQL	117,397
raw_azure_barnebys_lots	Lot data from Azure	Azure MySQL	105,215
raw_auction_house	Auction house reference data	Manual upload	42
raw_skeleton_pricing	Fee structure per AH	Manual upload	34
raw_currency_eur	EUR exchange rates	Manual upload	4

Output tables:

Table	Description	Record Count

proc_skeleton_auctions_with_enteredbids	Bids matched to EnteredBid with WebUserid & source	886,037
proc_skeleton_auctions_with_winning	Bids with winning flags & commissions	848,311
ana_lot_price_tiers	Lot level price tier classification based on max bid price distribution	157,669
ana_bids_with_price_tier	All bids records with price tier info	886,037
ana_winning_bids_with_price_tier	All winning bids with price tier info	848,311
ana_barnebys_increment	Lot-level Barnebys price increment	148,998
ana_funnel_olsens_exclued67	Funnel metrics by user source	2,299
ana_skeleton_fee	Monthly Skeleton fee per AH	378

4. Python Extraction Layer

Overview

The extraction layer automates data retrieval from three external databases and loads them into BigQuery. It consists of four modules:

File	Purpose
config.py	Date ranges, auction house list, database connection parameters
extract.py	Extraction functions for each data source
load.py	BigQuery loader
main.py	Pipeline orchestrator, runs extraction and SQL processing in order

Extraction methods by source

Skeleton SQL Server (Azure AD Authentication)

Each of the 34 auction houses has its own SQL Server database. main.py loops through all houses sequentially, extracting two queries per house:

- SQL_AUCTIONS_TEMPLATE: auction and inventory records
- SQL_ENTEREDBIDS_TEMPLATE: all entered bid records

Authentication uses Azure AD token via AzureCliCredential. Requires az.login before running. Loading strategy: first auction house uses WRITE_TRUNCATE (overwrite), subsequent houses use WRITE_APPEND to build up the unified table.

AWS MySQL (SSH Tunnel)

Lot data is extracted via an SSH tunnel to the AWS server using a .pem key file. All 34 auction houses are queried in a single call, filtered by auction_house_id and date range.

Azure MySQL (Direct Connection)

Lot data from June 2025 onwards is extracted via direct SSL connection to Azure MySQL. Also a single query across all 34 auction houses.

BigQuery loading

All data loaded via **load.py** using **google-cloud-bigquery**. The default model is WRITE_TRUNCATE (full refresh). Skeleton data uses WRITE_APPEND per auction houses as described above.

SQL Processing

After all raw data is loaded, **main.py** runs the SQL scripts in order via the BigQuery Python client:

- update_raw_bite_bids.sql
- raw_bbys_lots.sql
- raw_bite_bids_clean.sql
- proc_skeleton_auctions_with_bite_bids.sql
- proc_skeleton_auctions_with_enteredbids.sql
- proc_skeleton_auctions_with_winning.sql
- ana_lot_price_tiers.sql
- ana_bids_with_price_tier.sql
- ana_winning_bids_with_price_tier.sql
- ana_barnebys_increment.sql
- ana_funnel_all_stats.sql
- ana_funnel_excluded67_unpivoted.sql
- ana_funnel_excluded67.sql
- ana_skeleton_fee.sql

If any SQL script fails, the pipeline stops and logs the error. Fix the issue and re-run it.

Configuration

All parameters are defined in **config.py.** To update date ranges or add/remove auction houses, edit only **config.py**:

```
CONFIG = {
    "bq_project": "barnebys-skeleton",
    "bq_dataset": "42ah",
    "skeleton_start": "2025-01-01",
    "skeleton_end":   "2026-01-01",
```

```
  "bbys_aws_start": "2024-11-01",
  "bbys_aws_end":   "2025-05-31",
  "bbys_azure_start": "2025-06-01",
  "bbys_azure_end":   "2025-12-31",
}
```

To add a new auction house, add an entry to SKELETON_HOUSES and include the ah_id in AH_IDS.

5. Data Processing Logic

Core SQL processing logic is inherited from the Olséns pilot. See [Olséns Pilot Technical Documentation] for full detail on the following:

- Lot deduplication (AWS + Azure merge)
- Bite bids deduplication
- EnteredBid matching (Strict + Relaxed)
- Winning record handling
- Source inference
- Winning sign assignment
- User source classification (user_source_byfirstbid)
- Price tier classification (Low / Mid / High)

This section documents logic that is new or modified in the 34AH pipeline.

Currency Fix (Update_raw_bite_bids.sql)

Two auction houses have incorrect currency values in the raw_Bite_bids table and are corrected before processing:

Auction House	programid	Correction
Bastionen	3862	currency set to DKK
Dahlstroms	3898	currency set to EUR

This runs as the first SQL step to ensure all downstream EUR conversions are correct.

Funnel Analytics

This pipeline splits it into three layers to manage complexity across 34 auction houses:

Layer 1: ana_funnel_all_stats

Wide table aggregating all metrics by auction_house_id + user_source. Includes both per-AH rows and a Total row per source for cross-AH aggregation. All metrics are pre-calculated here, including:

- Funnel metrics: clicks, registered, bidders, bid, winners
- Behavioral metrics: avg_bids_per_user, users_bid_more_than_one_lot, unique_underbidders
- Revenue metrics: total_winning_value, total_commission, barnebys_increment
- EUR equivalents for all value fields

Layer 2: ana_funnel_excluded67_unpivoted

Unpivots ana_funnel_all_stats from wide to long format, one row per auction_house_id + user_source + event_type. This format is required for Looker Studio compatibility.

Note: clicks values are only populated for barnebys and Total user sources, as click data is not broken down by source in Bite.

Layer 3: ana_funnel_excluded67

Final presentation layer. Adds:

- Auction house names (joined from raw_auction_house)
- event_order for consistent sorting in dashboard
- percentage_of_total: Barnebys share of each metric vs total
- percentage_of_clicks: conversion rate from clicks for key funnel steps

Skeleton Fee Calculation (ana_skeleton_fee)

Calculates the monthly fee each auction house pays to Skeleton, based on three pricing models:

Type	Formula	Description
Fixed	fixed_fee_eur	Flat monthly fee regardless of sales volume
Percent	month_hammeredprice_eur × rate	Percentage of total monthly winning value
Hybrid	MAX (month_hammeredprice_eur × rate, min_fee_eur)	Higher of percent-based fee or minimum fee

All fees are converted to EUR using 2025 annual average exchange rates from raw_currency_eur.

Missing Month Handling

For Fixed and Hybrid auction houses, months with no auction activity still require a fee row (Fixed fee is still owed; Hybrid uses min_fee_eur). Missing months are detected and filled with zero-value rows and the appropriate fee:

- Fixed AHs: full 12 months required
- Hybrid AHs (except MyntAuktioner 2659): full 12 months required
- MyntAuktioner (2659): only March, May, July required fill
- Percent AHs: no fill needed (zero sales = zero fee)

bbys_tier Classification

Each auction house is assigned a performance tier based on total barnebys_increment_eur across 2025 (excluding June–July), calculated as a window function over all months:

Tier	Condition
High	Annual barnebys_increment_eur ≥ 100,000
Mid	Annual barnebys_increment_eur 10,000 – 99,999
Low	Annual barnebys_increment_eur < 10,000

This tier is used in the dashboard to prioritize which auction houses Barnebys should focus traffic on.

6. Metrics Definition

Core metric definitions are inherited from the Olséns pilot. See [Olséns Pilot Technical Documentation] for full detail on funnel metrics, conversion rates, revenue metrics, Barnebys incremental value, bidding behavior metrics, and price tier distribution.

This section documents metrics that are new or extended in the 34AH pipeline.

EUR Conversion

All value metrics have a corresponding **_eur** field for cross auction house comparison. Conversion uses 2025 annual average exchange rates from raw_currency_eur. This allows direct comparison across auction houses operating in SEK, DKK, NOK, and EUR.

7. Known Issues & Limitations

Core limitations inherited from the Olséns pilot (fingerprint reliability, incomplete Bite coverage, source inference reliability, InventoryId extraction consistency, Skeleton settlement timing) still apply.

This section documents issues specific to the 34AH pipeline.

Dataset Naming

The BigQuery dataset and local folder are named **42ah** for historical reasons, reflecting an earlier scope estimate. The actual pipeline covers 34 auction houses. This inconsistency is cosmetic only and does not affect data or logic.

Missing Month Handling

For Fixed and Hybrid auction houses, months with no auction activity are filled with zero-value rows in ana_skeleton_fee. This ensures fee obligations are still recorded even in inactive months. The fill logic is hardcoded for specific auction houses (e.g. MyntAuktioner 2659 requires only March, May, July). Any new auction house onboarded mid-year may require custom month handling.

Currency Hardcoding

Two auction houses have incorrect currency values in the raw Bite data, corrected via update_raw_bite_bids.sql. If new auction houses are onboarded with currency issues, this script must be updated manually.

Azure AD Token Expiry

Skeleton extraction loops through 34 databases sequentially, which can take significant time. Azure AD tokens may expire mid-loop. **main.py** handles this with automatic retry logic, but if az.login fails or requires MFA interaction, the pipeline will pause and require manual intervention.

8. SQL Scripts Reference

Script Inventory

Script	Layer	Description
bite_raw_clicks.sql	Raw	Extract clicks records from Bite
bite_raw_registrations.sql	Raw	Extract registration events from Bite
bite_raw_bids.sql	Raw	Extract bid events from Bite

skeleton_raw_auctions.sql	Raw	Extract auction & inventoryId records & hammeredprice from Skeleton
skeleton_raw_enteredbids.sql	Raw	Extract enteredbid records from Skeleton
aws_barnebys_raw_lots.sql	Raw	Extract lot data from AWS
azure_barnebys_raw_lots.sql	Raw	Extract lot data from Azure
update_raw_bite_bids.sql	Raw	Fix currency for Bastionen (DKK) and Dahlstroms (EUR)
raw_bbys_lots.sql	Raw	Merge AWS + Azure lot data, deduplicate by latest created timestamp
raw_bite_bids_clean.sql	Raw	Deduplicate Bite bids by inventoryId + value, keep earliest timestamp
proc_skeleton_auctions_with_bite_bids.sql	Processed	Match Bite bids to Skeleton auctions
proc_skeleton_auctions_with_enteredbids.sql	Processed	Match to EnteredBid, infer source (All bids records)
proc_skeleton_auctions_with_winning.sql	Processed	Assign winning flags and commissions (All winning bids records)
ana_lot_price_tiers.sql	Analytics	Price tier classification per lot
ana_bids_with_price_tier.sql	Analytics	All bids enriched with price tier
ana_winning_bids_with_price_tier.sql	Analytics	All winning bids enriched with price tier
ana_barnebys_increment.sql	Analytics	Lot-level Barnebys price increment
ana_funnel_all_stats.sql	Analytics	Wide table: all metrics by auction house + user source
ana_funnel_excluded67_unpivoted.sql	Analytics	Unpivot ana_funnel_all_stats to long format
ana_funnel_olsens_exclued67.sql	Analytics	Final funnel with AH names, ordering, percentages
ana_skeleton_fee.sql	Analytics	Monthly Skeleton fee per AH, missing month fill, bbys_tier

Triggered automatically by main.py after raw data load.

Dependencies

```
raw_bite_bids
    ↓
update_raw_bite_bids (currency fix)
    ↓
raw_bite_bids_clean
    ↓
proc_skeleton_auctions_with_bite_bids ← raw_skeleton_auctions
                                      ← raw_bbys_lots (← raw_bbys_aws_lots + raw_bbys_azure_lots)
    ↓
proc_skeleton_auctions_with_enteredbids ← raw_skeleton_enteredbids
    ↓
```

```
proc_skeleton_auctions_with_winning ← raw_skeleton_pricing
                                    ← raw_currency_eur
    ↓
    ├── ana_lot_price_tiers
    │           ↓
    │     ana_bids_with_price_tier
    │           ↓
    │     ana_winning_bids_with_price_tier
    │
    ├── ana_barnebys_increment
    │
    ├── ana_funnel_all_stats
    │           ↓
    │     ana_funnel_excluded67_unpivoted
    │           ↓
    │     ana_funnel_excluded67 ← raw_auction_house
    │
    └── ana_skeleton_fee ← raw_skeleton_pricing
                         ← raw_currency_eur
                         ← raw_auction_house
```

9. Long-term Strategy and Scalability

Core pipeline design principles and multi-auction house database strategy are inherited from the Olséns pilot. This section documents scalability considerations specific to the 34AH pipeline.

Adding New Auction Houses

To onboard a new auction house, only config.py needs to be updated:

```python
# Add to SKELETON_HOUSES
{"ah_id": "XXXX", "database": "NewAuctionHouseName"}
# Add to AH_IDS
AH_IDS = [..., 'XXXX']
```

Additionally verify

Check	Action
Bite tracking active	Confirm Bite's **programid** is aligned with auction_house_id Check the **source** is in the right track
Currency correct	Check raw_bite_bids after extraction Add fix to update_raw_bite_bids.sql if needed

Pricing model confirmed	Add row to raw_skeleton_pricing with correct type, rate, and fee
Missing month handling	Check if AH has irregular auction schedule requiring custom month fill in ana_skeleton_fee
InventoryId URL structure	Verify URL format matches existing parsing logic

Re-running the Pipeline

To update data for a new period:

- Update date ranges in **config.py**
- Update raw_currency_eur with new period average rates if currency changed
- Run **python main.py**

All tables use CREATE OR REPLACE — full refresh on every run.

Recommended Next Steps

Migrate BigQuery dataset from personal account to company BigQuery project to ensure data ownership, access control, and continuity. Automate Bite data extraction (clicks, registrations, bids) — currently requires manual BigQuery queries, integrating into extract.py and main.py would make the pipeline more automated end-to-end.

10. Dashboard

Overview

The analysis results are visualized in a Looker Studio dashboard, providing an interactive view of Barnebys user performance across 34 Skeleton auction houses for 2025..

Access: https://lookerstudio.google.com/s/tucgQNWlVbc

Data excludes June and July 2025 due to source tracking issues.

Dashboard Pages

Page	Title	Description
1	Barnebys Users Performance	Funnel overview (clicks → registered → bidders → winners), engagement metrics, winning value and commission breakdown.
2	Bidding Behavior – Price Formation & Incremental Value	Price tier analysis, Barnebys incremental value

3	Category Segment	Bids share, winning value, revenue concentration by category (Barnebys category)
4	Revenue Impact & User Economic Value	Skeleton fee analysis by pricing model (Fixed / Percent / Hybrid), Strategy mapping
5	Full funnel data table	Filterable by auction house

How to Refresh

- Re-run the full pipeline: python main.py
- Open Looker Studio dashboard
- Click **Refresh data** on the data source connected to **barnebys-skeleton.42ah**

11. Appendix

34 Auction House List

AH ID	Name	Database	Pricing Type
3915	Aros frimärken	Arosfrimarken	Fixed
3952	Auksjonshallen	Auksjonshallen	Fixed
3862	Bastionen	Bastionen	Percent
3900	Carlsson Ring Auktioner	CarlssonRing	Percent
3902	Conap Auktioner	Conap	Fixed
3898	Dahlströms Rare Prints	Dahlstroms	Fixed
44	Dalarnas Auktionsbyrå	Dalarnas	Hybrid
53	Dyrgripen	Dyrgripen	Hybrid
230	Eek Auktionsforretning AS	EekAuksjon	Fixed
3922	Gun garage	GunGarage	Hybrid
90	Ingelmark	Ingelmark	Fixed
3768	Jämtloppan	Jamtloppan	Fixed
3031	Kanonauktioner	kanonauktioner	Percent
3601	Karl-Johan Auktioner	Karljohan	Percent
3756	Karlssons Auktioner	Karlssons	Fixed

2949	Knutson & Bloom	KnutsonBloom	Fixed
3849	Laholms HAP	LaholmsHAP	Fixed
107	LP-Foto	Lpfoto	Fixed
2659	Myntauktioner i Sverige AB	MyntAuktioner	Hybrid
3925	Mälardalen Auktioner	Malardalen	Fixed
72	Nya Hallands Auktionsverk	NyaHallands	Hybrid
3687	Olséns Auktioner	Olsens	Percent
125	Probus	Probus	Hybrid
3865	Rekomo	Rekomo	Hybrid
3950	Ronald McDonald hus	McDonald	Fixed
3723	Snapphane Auktioner	Snapphane	Percent
3923	Store Mæla Gård Antikk	StoreMaele	Fixed
3904	Sydklippet	Sydklippet	Hybrid
283	Södersens Auktioner AB	Sodersen	Hybrid
3764	Sølvesen Myhre Berntsen	Solvesen	Fixed
3722	Upplands Auktionsverk	Upplands	Hybrid
3869	Wunderkammer	Wunderkammer	Hybrid
3663	Ystads auktioner	Ystads	Fixed
3916	Östlandet	Ostlandet	Fixed

Contact

Danqing

danqing.yao@barnebys.com

yaodq0709@gmai.com